A Project Report on

Blockchain Enabled Health Monitoring System

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Engineering

in

Computer Science and Engineering - Data Science

by

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Declaration

We declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, We have adequately cited and referenced the original sources. We also declare that We have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

This project explores the development of a secure and scalable smart health monitoring system that combines Internet of Things (IoT) devices with blockchain technology to enhance patient care and data management. The system uses a wristband equipped with sensors to monitor real-time health metrics like heart rate, pulse, and body temperature. IoT devices collect and transmit data, which is securely stored on a blockchain, ensuring data integrity and preventing unauthorized access. By addressing key challenges such as data privacy, security, and interoperability, the system aims to improve healthcare outcomes, enhance patient engagement, and ensure efficient, secure data handling. The Blockchain-Enabled Health Monitoring System merges IoT technology and blockchain to create a secure, scalable platform for real-time patient health monitoring and data management. Using a wristband equipped with sensors, the system continuously tracks vital health metrics such as heart rate, pulse, and body temperature, transmitting this data securely via IoT devices. Blockchain plays a critical role in ensuring data integrity, privacy, and security by storing health data in a decentralized, tamper-proof ledger, making unauthorized access or data manipulation nearly impossible. The system addresses key healthcare challenges such as data privacy, interoperability, and transparency, allowing seamless data sharing between healthcare providers while maintaining a clear audit trail of data access. Additionally, smart contracts enforce granular access control, ensuring that only authorized personnel can interact with sensitive patient data. This real-time health tracking not only improves patient care and preventive health management but also fosters better engagement by empowering patients with access to their own health information through user-friendly mobile and web applications. By integrating blockchain with IoT, the system enhances security and trust in healthcare data handling, offering a robust solution for modern healthcare needs.

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IOT:	Internet Of Things
ML:	Machine Learning
BPM:	Beat Per Minute
ECG:	Electro Cardio Graph
AI:	Artificial Intligence
EHRs:	Electronic Health Records

Chapter 1

Introduction

In the rapidly evolving digital era, healthcare technology has seen significant advancements, particularly in the field of real-time health monitoring. Traditional healthcare systems often face challenges such as delayed diagnosis, inefficient patient monitoring, and a lack of continuous data collection, leading to increased health risks. The integration of the Internet of Things (IoT) with machine learning (ML) presents an innovative approach to overcoming these challenges. This project focuses on developing a real-time IoT-based health monitoring system that enables continuous tracking of vital health parameters, allowing for timely interventions and better patient outcomes. The proposed system utilizes an ESP32 microcontroller, which is integrated with MAX30100 (pulse oximeter) and DHT11 (temperature & humidity sensor) to collect essential health data, including heart rate, oxygen saturation (SpO_2), temperature, and humidity. This data is transmitted to Firebase, a cloud-based platform that ensures secure storage and real-time accessibility. Furthermore, a machine learning model is employed to analyze the collected data and predict potential health risks. The final results are displayed through a mobile application, providing users with real-time insights and alerts about their health status. The primary objective of this project is to design an efficient IoT and ML-driven health monitoring system that offers real-time data collection, cloud-based storage, and predictive analysis to help individuals monitor their health more effectively. The system is particularly beneficial for elderly patients and individuals with chronic conditions who require continuous health tracking. The scope of the project extends beyond personal health monitoring, as it can also be utilized in hospitals, remote healthcare services, and fitness tracking applications. Additionally, this system lays the foundation for future advancements, including AI-powered diagnostics and wearable technology integration, making it a scalable and adaptable solution for modern healthcare needs. This project contributes significantly to the field of IoT-based healthcare by integrating real-time health monitoring with predictive analytics. The use of machine learning enhances the accuracy of early diagnosis, reducing the chances of critical health deterioration. By leveraging Firebase for seamless data transmission and cloud storage, the system ensures efficient remote health monitoring while maintaining data security and accessibility. The mobile application interface enhances user experience by offering intuitive and real-time insights, allowing individuals to take proactive measures for their well-being. Ultimately, this project aims to bridge the gap between technology and healthcare, providing a smart, scalable, and efficient solution for modern health monitoring challenges.

1.1 Motivation

In today's fast-paced world, healthcare challenges such as delayed diagnosis, lack of continuous monitoring, and limited accessibility to medical facilities have highlighted the need for real-time health tracking solutions. Many individuals, especially the elderly and those with chronic illnesses, require continuous monitoring to detect potential health risks before they become critical. Traditional healthcare systems often fail to provide this level of real-time care, making it essential to integrate IoT technology, cloud computing, and machine learning for efficient and proactive health monitoring. This project is driven by the goal of developing a cost-effective, smart, and accessible health monitoring system that enables continuous tracking and predictive analysis of vital health parameters.

Using an ESP32-based IoT system with Firebase and machine learning algorithms, this project aims to bridge the gap between healthcare and technology, ensuring real-time data collection, secure storage, and intelligent health predictions. Unlike expensive commercial health monitoring solutions, this system is designed to be affordable and scalable, making it suitable for individuals, hospitals, and remote health services. The ultimate motivation is to empower users with real-time insights into their health, reduce hospital visits through early detection, and contribute to the advancement of smart healthcare solutions for better patient outcomes.

1.2 Problem Statement

In today's healthcare landscape, timely detection and continuous monitoring of vital health parameters remain a critical challenge, particularly for individuals suffering from chronic diseases, elderly patients, and those in remote areas with limited access to healthcare facilities. Traditional healthcare systems rely on periodic check-ups and hospital visits, which often lead to delayed diagnosis and ineffective health management. The lack of a real-time, accessible, and intelligent monitoring system increases the risk of severe health conditions going unnoticed, resulting in higher medical costs, emergency hospitalizations, and preventable complications.

Need for Real-Time Health Monitoring & Secure Data Handling: Many existing IoT-based healthcare systems fail to provide real-time monitoring due to limited infrastructure for processing and storing large volumes of health data in a scalable and secure manner. Most traditional systems rely on centralized databases, making them vulnerable to data breaches, unauthorized access, and tampering. The lack of interoperability among healthcare providers further complicates data sharing and analysis, reducing the efficiency of medical decision-making.

Integration Challenges in IoT-Based Health Systems: The fragmented nature of healthcare technology makes it difficult to integrate wearable IoT devices with cloud-based systems for continuous patient monitoring. Existing wearable devices often lack predictive analytics, secure data transmission, and cloud connectivity, making it difficult to detect potential health risks in advance. There is an urgent need for a secure, efficient, and scalable IoT-based health monitoring system that can provide real-time patient data while ensuring data privacy and security.

Ensuring Data Privacy & Security in Healthcare Systems: The increasing digitization of health records raises concerns about data privacy, security, and integrity. Sensitive patient information is at risk of being accessed by unauthorized entities due to the lack of proper encryption and security protocols in most existing systems. A blockchain-enabled system could provide tamper-proof data storage, cryptographic security, and smart contract-based access control, ensuring that only authorized personnel can interact with sensitive patient data.

Challenges in Interoperability & Remote Access to Health Data: Most healthcare providers use different platforms and systems that do not communicate effectively with one another, making it difficult to share, retrieve, and analyze patient data across various organizations. Additionally, patients often lack access to their own health data, making proactive health management difficult. A mobile application integrated with IoT, cloud storage, and machine learning-based predictive analytics could empower both patients and healthcare providers with real-time health insights and remote access to health records.

1.3 Objectives

To address these limitations, this project proposes the development of a real-time IoT-based health monitoring system that integrates ESP32 with MAX30100 (pulse oximeter) and DHT11 (temperature and humidity sensor) to continuously track key health metrics. By utilizing Firebase for cloud-based storage and machine learning algorithms for predictive analysis, the system will ensure accurate health monitoring, early risk detection, and remote accessibility through a dedicated mobile application. Additionally, blockchain technology will be implemented for decentralized, tamper-proof storage of health data, ensuring that patient information remains secure, immutable, and accessible only to authorized users. This project aims to bridge the gap between healthcare and technology, offering an efficient, affordable, and scalable solution for continuous health tracking and proactive patient care

- To develop an IoT-based health monitoring system using ESP32 with MAX30100 and DHT11 sensors to provide real-time tracking of vital health parameters, ensuring continuous monitoring and data transmission.

- To utilize Firebase cloud storage to securely store real-time health data, allowing for easy access, retrieval, and historical tracking. This will facilitate remote monitoring by healthcare providers and data-driven insights for patients.

- To implement a machine learning model to analyze health data trends and provide risk predictions for potential health issues. By applying predictive analytics, the system will help detect anomalies and patterns, enabling early intervention and preventative healthcare measures.

- To ensure secure data encryption and privacy controls by implementing robust encryption protocols to protect patient health data from unauthorized access. Additionally, use blockchain technology for tamper-proof storage and smart contract-based access control, allowing only authorized healthcare providers and patients to interact with sensitive medical data.

- To develop an intuitive mobile application that allows patients and healthcare providers to remotely access real-time and historical health data. The app will provide visual analytics, health insights, and alert notifications, empowering users to take proactive steps toward better health management.

- To improve healthcare efficiency through seamless data interoperability by enabling secure and efficient data exchange between healthcare providers. By integrating standardized data protocols and cloud-based APIs, the system will facilitate better communication and collaboration in patient care, reducing medical errors and improving overall healthcare outcomes.

1.4 Scope

- Personal Health Monitoring – The system enables individuals to continuously track their vital signs, such as heart rate, SpO2 levels, and body temperature, using an IoT-based wearable device. Real-time alerts notify users of any abnormalities, promoting proactive health management and reducing dependency on periodic medical check-ups.

- Hospital & Clinical Use – The system facilitates remote patient monitoring for doctors and healthcare professionals, providing real-time access to patient health data. Historical data stored in Firebase allows for better diagnosis, treatment planning, and personalized healthcare interventions.

- Remote & Rural Healthcare – Designed for accessibility in rural and resource-limited areas, the system ensures that individuals without immediate access to healthcare facilities can monitor their health conditions effectively. The integration of cloud storage and a mobile app allows for remote consultation and emergency alerts.

- Data Security and Privacy – The system employs robust encryption protocols and blockchain technology to ensure the confidentiality and integrity of sensitive patient data. By leveraging blockchain's decentralized ledger, health records remain tamper-proof and accessible only to authorized healthcare providers and patients.

- Mobile and Web-Based Accessibility – A user-friendly mobile application provides patients and healthcare professionals with real-time access to health metrics, historical trends, and AI-driven health insights. The app serves as a seamless interface for monitoring, analyzing, and sharing critical health information.

- Scalability & Future Expansion – The project is designed for future scalability, allowing integration with AI-based predictive analytics, additional biomedical sensors, and telemedicine services. This ensures continuous improvement in healthcare monitoring, making the system adaptable to future technological advancements.

The scope of this IoT-based Health Monitoring System encompasses the design, development, and deployment of an integrated health-tracking solution that enhances patient care through real-time monitoring, secure data management, and intuitive user interfaces. The system integrates an ESP32 microcontroller with MAX30100 (pulse oximeter) and DHT11 (temperature and humidity sensor) to monitor key health parameters continuously. By utilizing cloud-based storage in Firebase, the system ensures seamless data access for patients and healthcare providers, enabling timely medical interventions and improving healthcare outcomes. A critical component of the project is ensuring data security and privacy. The system employs end-to-end encryption for secure data transmission and blockchain technology for tamper-proof storage. This decentralized approach prevents unauthorized access and enhances data integrity, ensuring that health records remain protected from breaches and modifications.

Chapter 2

Literature Review

Blockchain-enabled health monitoring systems using IoT devices offer enhanced security, pri vacy, and interoperability for healthcare data. IoT devices collect real-time patient informa tion, but concerns over data security and privacy persist. Blockchain, with its decentralized and tamper-proof features, ensures secure transmission and integrity of health data. The integration of blockchain and IoT allows for improved patient control over data access and enhances trust among healthcare stakeholders. However, challenges such as scalability, la tency, and regulatory concerns need to be addressed for widespread adoption in healthcare. Research continues to explore solutions like hybrid blockchain models and better consensus mechanisms.

Deependra Sinha et al. (2021) [13] developed a telemedicine-based health monitoring sys tem using IoT technologies, including pulse sensors, an Arduino UNO microcontroller, and the BLYNK App for real-time data transfer, though it is limited by heavy hardware and restricted mobility. Maroua Ahmid et al. (2020)[2] introduced a secure health monitoring system that employs multi-agent systems and cryptography, collecting heart rate data from Raspberry Pi 3 B+ with real-time notifications. However, its reliance on internet connectiv ity and agent complexity limits its effectiveness, particularly in remote areas. Similarly, Dr. Anil Ramachandran Nair et al. (2019)[11] proposed an IoT-based system using sensors for monitoring physiological parameters like heart rate and ECG, facing issues with managing large-scale data, power dependency, and limited internet accessibility in rural regions. Ra jdeep Kumar Nath and Himanshu Thapliyal (2021)[7] created a wearable wristband for stress detection using machine learning on physiological signals, though its accuracy is hindered by environmental factors and potential discomfort from wearing sensors. Other research includes Rajesh TM et al.'s (2022)[9] IoT-based health monitoring sys tem using homomorphic encryption for secure data transmission, which is limited by slow processing and Bluetooth range issues. Apratim Shrivastav et al. (2022)[12] introduced an optimized algorithm for step count estimation but noted challenges in noisy environments. Kaveri Ramesh Dabhade and Kiran Suresh Mulik (2020)[4] developed a smart health band that monitors vital signs but struggles with internet dependency and sensor accuracy. IoT enabled systems with deep learning models, such as those proposed by Xingdong Wu et al. (2021)[14] and Shaikh Anowarul Fattah et al. (2017)[5], focus on real-time physiological data analysis but face computational challenges and model complexity. Rangga Adi F. and Bambang Guruh I. (2019)[6] developed an IoT-based system for remote monitoring of heart rate and body temperature. The system captures ECG signals using electrodes and processes them through amplifiers and filters to eliminate noise, while temperature readings are taken using the DS18B20 sensor. However, the accuracy of the readings is significantly affected by patient movement, introducing a notable error margin. Alaa Awad Abdellatif (2018)[1] proposed a secure healthcare system using blockchain technology. By leveraging blockchain's decentralized and tamper-resistant ledger, it ensures secure data sharing with encryption for privacy, while smart contracts automate healthcare processes. This approach enhances patient data integrity and system trustworthiness, but faces drawbacks such as potential scalability issues, high energy consumption, and integration challenges with existing healthcare systems. Suparat Yongjoh, Chakchai So-in, and Peerapol Kompunt S. (2021)[15] explored the use of blockchain with decentralized storage, smart contracts, and encryption for secure patient data sharing across healthcare applications. While this approach aims for immutable and secure data exchange, challenges include complex integration with current healthcare systems, scalability issues, and the energy demands of consensus mechanisms. Wafaa A. N. A. Al-Nbhany and Ammar T. Zahary (2023)[3] examined blockchain-IoT applications in healthcare, particularly for real-time health monitoring through IoT sensors and machine learning models. Their work focuses on using blockchain for decentralized data storage and smart contracts to ensure data integrity. Key challenges include high latency in processing real-time data, device interoperability issues, and energy consumption concerns that can impact scalability. Rucha Patel and Vedvyas Dwivedi (2024)[8] reviewed the integration of blockchain with IoT for smart health monitoring, emphasizing the technology's role in secure data exchange and patient privacy. The study highlights significant challenges such as scalability, high energy use, complex integration, and regulatory barriers, which can hinder the widespread adoption of such systems. Prayag Tiwari and Deepak Gupta (2023)[10] introduced an anonymous IoT-based e health monitoring system that relies on edge computing for processing data. This approach aims to address privacy concerns and security risks. However, it faces challenges like limited IoT device capacity, dependence on edge nodes, and the complexity of integrating IoT in medical technology (IoMT) with existing healthcare infrastructures.

Sr. No	Title	Author(s)	Year	Methodology	Drawback
1	Heart Rate Monitoring System[1]	Deependra Sinha, Tapas Gupta, Shraddha Verma, Sweety Verma,Vijaylaxmi	2021	The research focuses on a telemedicine-based patient health monitoring system using IoT technologies. It uses pulse sensors,Arduino UNO microcontrollers, and the BLYNK App for real-time data transfer and visualization. The system uses a wired connection for remote monitoring and aims for simplicity.	The research highlights limitations of a device, including heavy hardware, slow data transfer, and limited mobility due to its tethered nature, which limits the patient's ability to move freely in dynamic environments
2	An Intelligent and Secure Health Monitoring System Based on Agent[2]	Maroua Ahmid, Okba Kazar, Saber Benharzallah, Laid Kahloul, Abdelhak Merizig	2020	The paper presents a secure IoT-based health monitoring system that uses multi-agent systems and advanced cryptography for real-time monitoring. It features an agent-based architecture, collecting heart rate data from Raspberry Pi 3 B+ with a pulse sensor, and uploading it to the cloud for secure storage and real-time access. The system also sends real-time notifications for abnormal heart rate levels.	The advanced system has limitations,including reliance on constant internet connectivity, hardware limitations due to Raspberry Pi and pulse sensors, and agent complexity, which may limit monitoring scope and management over time, especially in rural or remote areas.
3	IoT-based Secure Healthcare Monitoring System[3]	Dr. Anil Ramachandran Nair, Dr. R. K. Sharma, Naveen	2019	The research presents an IoT-based healthcare monitoring system that uses sensors and a cloud-based platform to monitor patient physiological parameters. The system uses sensors like heart rate, body temperature, ECG, and blood pressure, and a Raspberry Pi for processing data.	The system faces challenges in managing large volumes of real-time healthcare data, limited accessibility due to poor internetcoverage, scalability issues for large hospitals, and power dependency due to its dependence on a continuous power supply, which can be problematic in unstable electricity areas.

Table 2.1: Comparative Analysis of Literature Survey

Sr. No	Title	Author(s)	Year	Methodology	Drawback
4	Wearable Health Monitoring System for Older Adults in a Smart Home Environment	Rajdeep Kumar Nath, Himanshu Thapliyal	2021	The methodology involves using wearable sensors (EDA, PPG, and ST) in a wristband to monitor stress, with salivary cortisol as the stress biomarker. Machine learning models were trained using features from these physiological signals for stress detection.	Some drawbacks of the system include the reliance on wearable sensors, which may be uncomfortable or prone to user non-compliance over long periods. The blood pressure model, based solely on PPG signals, may lack the accuracy of multi-sensor approaches like ECG. Additionally, real-time data streaming and processing could introduce delays, and the system's efficacy may be limited by environmental factors affecting sensor accuracy.
5	Secure Remote Health Monitoring System and assessment using IOT	Rajesh TM, Tina Babu, Rekha R Nair, Nivedha S	2022	The system uses embedded sensors (e.g., temperature and heart rate) connected to an Arduino Nano microcontroller to collect health data, which is transmitted via Bluetooth to a cloud server for real-time analysis. Homomorphic encryption ensures secure data transfer, and the system categorizes health status as critical, normal, or dangerous.	The system faces challenges like limited Bluetooth range, which may cause connectivity issues in larger areas. The use of homomorphic encryption can slow down real-time data processing, and the reliance on external sensors adds hardware complexity and maintenance costs. Additionally, sensor accuracy could be impacted by environmental factors, affecting the reliability of health assessments.
6	Optimised Algorithm for Step Count Estimation Using Sensor Data from Smartphones and Wearables	Apratim Shrivastav, Utkarsh Kuchhal, Samarth Singhal, Tanmay Jain, Divyashikha Sethia, Vidushi Chaudhary	2022	The methodology proposes an optimized algorithm for step count estimation using accelerometer data by normalizing acceleration across the x, y, and z axes. A low-pass Butterworth filter reduces noise, and peak detection identifies steps while avoiding false peaks. The algorithm is highly accurate, outperforming machine learning and deep learning models like SVM and CNN in step count estimation.	The algorithm's peak detection may struggle with irregular walking patterns and abrupt movements, leading to missed steps or over-counting in noisy environments.

Table 2.2: Comparative Analysis of Literature Survey

Sr. No	Title	Author(s)	Year	Methodology	Drawback
7	IOT based Wearable Smart Health Band Assistance	Kaveri Ramesh Dabhade, Kiran Suresh Mulik	2020	The methodology involves using sensors (pulse, temperature, and alcohol) connected to an Arduino Nano to monitor vital signs like heart rate, body temperature, and alcohol levels. Data is processed and transmitted via a Bolt IoT Wi-Fi module to the cloud for real-time monitoring through a smartphone app.	The system's drawbacks include its reliance on constant internet connectivity, which may limit functionality in areas with poor coverage. The sensors used, particularly for alcohol detection, may not provide highly accurate medical-grade results.
8	Internet of things-enabled real-time health monitoring system using deep learning	Xingdong Wu, Chao Liu, Lijun Wang, Muhammad Bilal	2021	The methodology involves using IoT-enabled wearable devices to collect real-time physiological data from athletes, which is transmitted via a wireless relay network for analysis. Deep learning algorithms, specifically deep neural networks (DNN), process the data to detect health issues.	The drawbacks of the system include its sensitivity to the number of neurons in the DNN, which can lead to issues like underfitting or overfitting. The deep learning model's complexity increases computational costs, and challenges like vanishing gradients may arise with deeper networks.
9	Wrist-Card: PPG Sensor based Wrist Wearable Unit for Low Cost Personalized Cardio Healthcare System	Shaikh Anowarul Fattah, Mohammad Mahinur Rahman, Nafis Mustakin, Mohammad Tariqul Islam, Asir Intisar Khan, Celia Shahnaz	2017	The methodology involves using IoT-enabled wearable devices to collect real-time physiological data from athletes, which is transmitted via a wireless network for analysis. Deep neural networks (DNN) process this data to detect health risks.	The main drawbacks include the deep learning model's sensitivity to neuron numbers, leading to potential underfitting or overfitting. The system's complexity also increases computational costs, and issues like vanishing gradients may arise with deeper networks.
10	Monitoring Heart Rate And Temperature Based On Internet Of Things	Rangga Adi F,Bambang Guruh I, Sumber	2019	The study uses an IoT-based system to monitor heart rate and body temperature remotely. ECG signals are captured via electrodes, processed through amplifiers and filters to remove noise, while body temperature is measured using the DS18B20 sensor.	The main drawbacks of the system include the significant error margin during patient movement, which affects the accuracy of heart rate and temperature readings.

Table 2.3: Comparative Analysis of Literature Survey

Sr. No	Title	Author(s)	Year	Methodology	Drawback
11	ssHealth: Toward Secure, Blockchain-Enabled Healthcare Systems	Alaa Awad Abdellatif	2018	The methodology leverages blockchain's decentralized, tamper-resistant ledger to ensure secure data sharing, encryption for privacy, and smart contracts to automate healthcare processes, enhancing patient data integrity and system trustworthiness.	Drawbacks include potential scalability issues, high energy consumption of blockchain networks, integration challenges with existing systems, and regulatory uncertainties regarding data privacy and compliance in healthcare.
12	Development of an Internet-of-Healthcare	SUPARAT YONGJOH,CHAKCHAI SO-IN and PEERAPOL KOMPUNT S	july 13, 2021	The methodology involves using blockchain technology with decentralized storage, smart contracts, and encryption to ensure secure, immutable patient data sharing across various healthcare applications and cloud systems.	TPotential drawbacks include integration complexity with existing healthcare systems, scalability issues, reliance on energy-intensive consensus mechanisms, and challenges in regulatory compliance and user adoption.
13	Blockchain-IoT Healthcare Applications and Trends	WAFAA A. N. A. ALNBHANY 1 , AMMAR T. ZAHARY	14 December 2023	wisthband fir monitoring using ml models , iot sensors Utilizes blockchain for secure, decentralized data storage, smart contracts for automated transactions, and IoT devices for real-time health monitoring and data integrity assurances.	Challenges include high latency in real-time data processing, interoperability issues between diverse IoT devices, and increased energy consumption impacting scalability and network efficiency.
14	A Review of Secure IoT based Smart Health Monitoring	Rucha Patel and Vedvyas Dwivedi	2024	Meteorology studies the atmosphere, focusing on weather processes like temperature, humidity, air pressure, wind, precipitation, clouds, storms, and forecasting using tools like satellites, radars, and computer models.	Blockchain-IoT healthcare faces challenges like scalability, high energy use, complex integration, privacy issues, storage limitations, regulatory hurdles, user adoption hesitance, and high initial costs.
15	An Anonymous IoT-Based E-Health Monitoring System	Prayag Tiwari , Deepak Gupta	2023	Meteorology branches include synoptic meteorology (weather patterns), dynamic meteorology (atmospheric motions), climatology (climate patterns), physical meteorology (clouds, precipitation), and atmospheric chemistry (pollutants, composition).	The project faces challenges like limited IoT device capacity, dependency on edge nodes, privacy concerns, data security risks, and complex integration of IoMT into existing systems.

Table 2.4: Comparative Analysis of Literature Survey

Chapter 3

Project Design

Project design refers to the structured planning and conceptualization phase of a project, where the overall architecture, system components, and workflow are defined. It outlines how the project will achieve its objectives by detailing the hardware setup, data flow, software architecture, and integration strategy. A well-thought-out design ensures smooth development, efficient performance, and scalability. In this project, the design focuses on integrating IoT-based health sensors with cloud storage and machine learning for real-time health monitoring and prediction.

3.1 Existing System

Another prominent approach focuses on IoT-enhanced remote health monitoring combined with Advanced Encryption Standard (AES) for data confidentiality. This system periodically (every 10 seconds) collects data from sensors such as DHT11 (temperature), ECG sensors, heart rate sensors, and blood pressure sensors and performs preprocessing techniques to handle missing values, outliers, and duplication. The preprocessed data is then sent to a Raspberry Pi 3 running Python-based algorithms, which compare current readings against normal physiological ranges. If any abnormality is detected, an alarm is triggered and a message is sent to medical personnel within a minute. Data is then encrypted using 128-bit AES and uploaded to a secure cloud database. Only authorized users with a matching passkey can decrypt and access the data.

The strength of this system lies in its emphasis on data accuracy, security, and timely response. It is particularly valuable in scenarios involving elderly patients who live alone, providing doctors with a reliable method to access patient information and respond to emergencies remotely. However, unlike the agent-based system, this design doesn't offer self-management or autonomous interaction among components, making it more dependent on centralized control.[2]

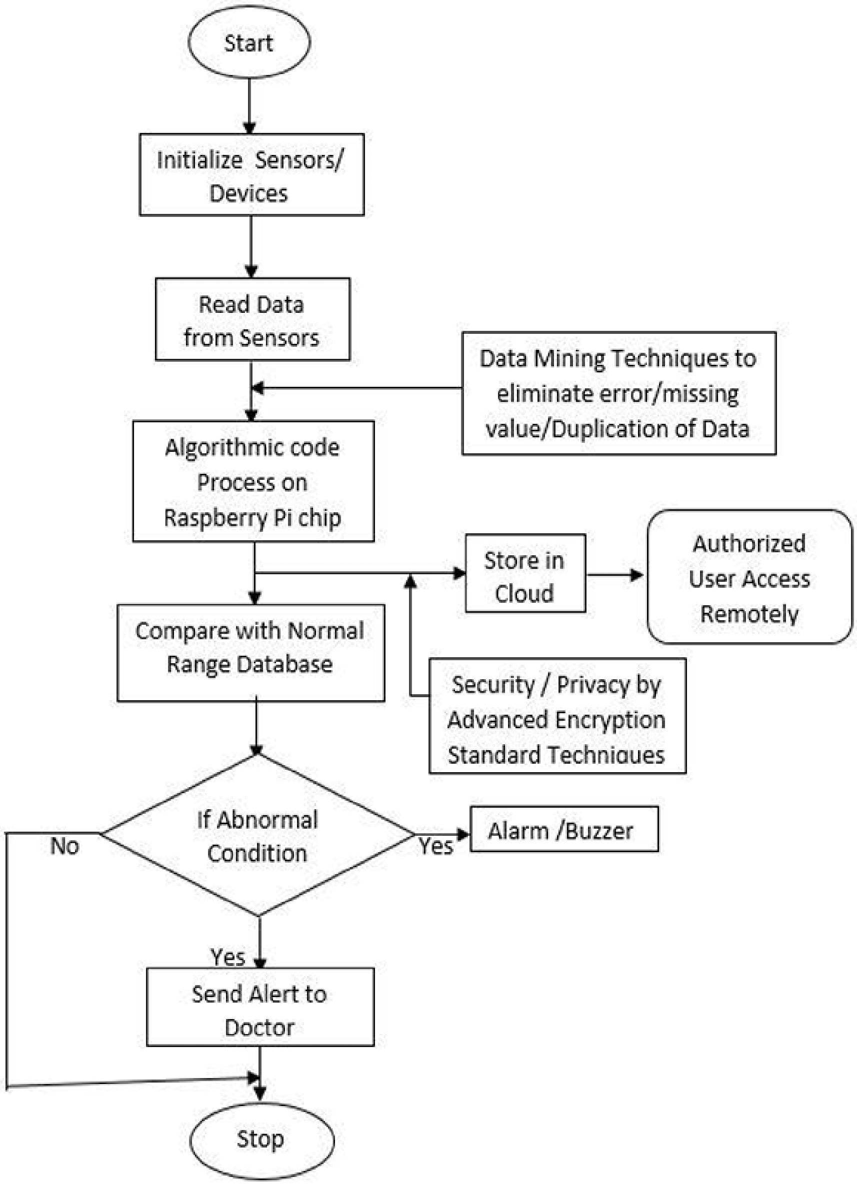

Figure 3.1: Existing System

One of the most advanced proposals in secure healthcare IoT systems is the Agent-Based Intelligent Health Monitoring Architecture (A-IoT), which incorporates agents into each layer of the IoT infrastructure. The system integrates cloud computing, real-time monitoring, and Elliptic Curve Cryptosystem ElGamal (ECC-ElGamal) encryption for lightweight but secure data transmission. It divides responsibilities across five layers—device agents, gateway agents, manager agents, analyzer agents, and user agents—each playing a role in decision-making and communication. The agent-based design helps improve autonomy, scalability, and self-adaptability of the system.

In their experimental setup, the authors used Raspberry Pi, ADS1115 ADC, and a Pulse Sensor Amped, which collected heart rate data. The information is processed by device agents and encrypted before being sent to the cloud. If an abnormal condition is detected (e.g., bradycardia or tachycardia), the system immediately triggers an alert via SMS to the doctor and patient's family. Real-time data is made available through a mobile application, ensuring that both caregivers and doctors can monitor health continuously. This approach showcases a comprehensive, secure, and responsive health monitoring system that outperforms conventional methods in flexibility and intelligence.

Despite its robust design, this system is complex and may incur higher development and maintenance costs due to its agent-based nature and encryption overhead. However, it clearly provides a scalable foundation for future smart healthcare systems with security and autonomy as core features.[3]

3.2 Proposed System

The Blockchain-Enabled IoT-Based Health Monitoring System is designed to offer real-time tracking of patient vitals, secure health data storage, predictive analytics, and enhanced privacy controls. This system integrates wearable IoT sensors, cloud storage, machine learning (ML), and blockchain security, creating a reliable and efficient healthcare monitoring platform. The primary objective is to ensure continuous health monitoring, timely alerts for abnormalities, and secure patient data handling, ultimately enhancing patient care and healthcare efficiency.

The system utilizes wearable IoT sensors such as the MAX30100 pulse oximeter for measuring heart rate and SpO_2 levels and the DHT11 sensor for monitoring body temperature and humidity. These sensors are connected to an ESP32 microcontroller, which processes and transmits the collected health data wirelessly. The real-time data is then sent to Firebase Cloud storage, ensuring secure and continuous health monitoring. Patients and healthcare providers can access historical health data, enabling trend analysis and informed decision-making.

The sensor data is stored in Firebase, a secure and scalable cloud storage system, which allows for real-time data retrieval and analysis. This ensures that doctors and patients can access health records anytime, facilitating remote health monitoring. The cloud-based approach enhances system efficiency by reducing data loss risks and providing easy access to medical history for diagnostics and treatment.

A machine learning (ML) model is implemented to analyze patient data trends and predict potential health risks. The model is trained using patient health records to detect anomalies such as irregular heart rate patterns or abnormal temperature variations. Algorithms such as Decision Trees, Random Forest, or LSTM (Long Short-Term Memory) are utilized for pattern recognition and trend analysis. When an anomaly is detected, the system automatically generates alerts and notifies both patients and healthcare providers for timely medical intervention.

To ensure data integrity, security, and privacy, the system integrates blockchain technology for storing patient records. Each health record is encrypted and stored on a decentralized ledger, making it tamper-proof and only accessible to authorized personnel. Blockchain's smart contract mechanism can also automate administrative processes such as insurance claims and patient consent management, ensuring a transparent and trustable healthcare system.

A dedicated mobile application is developed for both patients and healthcare providers. This app allows users to view real-time vitals, analyze historical trends, receive alerts, and engage in AI-driven health recommendations. The user-friendly interface enhances accessibility, ensuring that patients can monitor their health conditions from anywhere. Features such as teleconsultation, medication reminders, and emergency alerts further enhance the app's usability, promoting proactive healthcare management.

The overall workflow begins with IoT sensors collecting real-time health data, which is then transmitted to the Firebase cloud for storage and further analysis. The ML model processes the data, predicting potential health risks and alerting users to critical conditions. Meanwhile, blockchain technology secures the health records, preventing unauthorized access and ensuring patient privacy. The mobile application serves as the primary interface, providing patients and doctors with real-time insights and notifications.

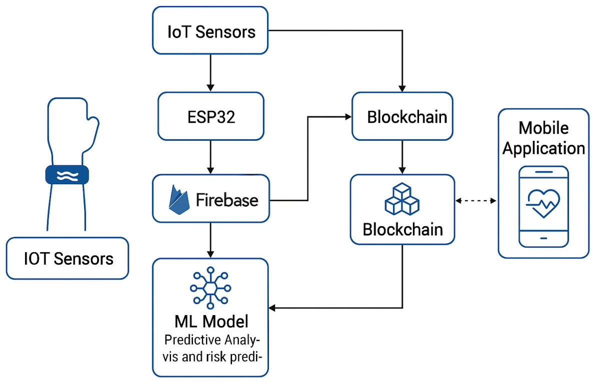

Figure 3.2: Proposed System

The proposed system creates a secure, AI-powered, and IoT-integrated health monitoring solution that enhances patient care and healthcare efficiency. By leveraging IoT for real-time tracking, ML for predictive analysis, and blockchain for security, this system ensures timely medical interventions, accurate diagnostics, and data protection. The combination of these cutting-edge technologies makes the system scalable, efficient, and suitable for modern healthcare applications, revolutionizing patient monitoring and proactive healthcare management.

3.2.1 Critical Components of System Architecture

- **IoT Sensors (MAX30100, DHT11):** Collect real-time health data such as pulse, oxygen levels, body temperature, and humidity. The sensors send this data to the ESP32 microcontroller.

- **ESP32 Microcontroller:** Acts as the central processor and controller for the system. It collects data from the sensors, processes it, and transmits it via secure communication protocols (e.g., Wi-Fi or Bluetooth) to the cloud (Firebase).

- **Firebase Cloud Storage:** Firebase is used for real-time storage and retrieval of health data. It ensures that the data is easily accessible by both the patients and healthcare providers for further analysis and monitoring.

- **Blockchain Technology:** The blockchain component ensures that the health data is stored in a decentralized, tamper-proof ledger. Blockchain helps maintain the integrity of sensitive data, making sure that only authorized parties can access it.

- **Machine Learning Model:** The ML model analyzes the health data to identify patterns and predict potential risks to patient health. It can trigger alerts when a concerning trend is detected (e.g., a sudden spike in heart rate).
- **Mobile Application:** The mobile application provides a user-friendly interface for both patients and healthcare providers to view real-time health data, receive alerts, and track trends.

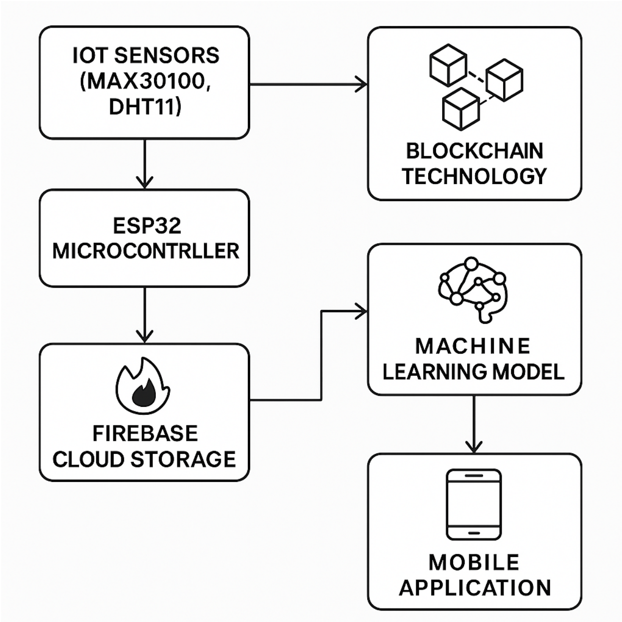

Figure 3.3: System Architecture

3.3 System Diagrams

System diagram represents the visual blueprint of the project's architecture, illustrating the interaction between various components and modules. It provides a clear overview of how data flows through the system from sensor data collection using hardware modules to storage in the cloud, followed by machine learning processing and output visualization. This diagram helps in understanding the system's functionality at a glance and ensures proper coordination between hardware, software, and network components in the IoT health monitoring framework.

3.3.1 Activity Diagram

The activity diagram illustrates the sequential flow of operations in the health monitoring system. It begins with the MAX30100 sensor collecting heart rate and SpO_2 data, which is then transmitted to the ESP32 microcontroller for processing. The ESP32 forwards this data to Firebase, where it is stored securely. The patient can then use a mobile application to request and view their health data, ensuring real-time monitoring.

Additionally, the ML model analyzes the stored data to detect potential health risks. If an anomaly is found, the system triggers an alert notification to the mobile app, informing the patient or caregiver. This structured workflow ensures that data is effectively collected, processed, analyzed, and delivered for timely health insights.

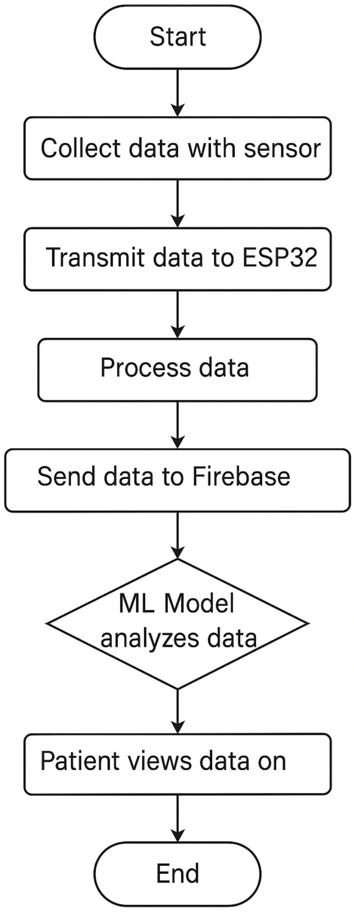

Figure 3.4: Activity Diagram

3.3.2 Use Case Diagram

The Health Monitoring System described in the diagram is a cutting-edge solution that utilizes wearable technology, real-time data transmission, secure storage, and mobile-based access to ensure continuous health monitoring. This system is designed to help patients and healthcare providers track vital health parameters, detect potential risks early, and enable timely medical intervention. By integrating IoT sensors, blockchain technology, and mobile applications, it ensures data security, accuracy, and accessibility in healthcare management.

The process begins with Metrics Collection, where the patient wears a wristband embedded with sensors that continuously monitor key physiological parameters such as heart rate, temperature, and blood pressure. These vital signs are recorded in real-time, ensuring that any sudden health fluctuations can be detected immediately. The advantage of continuous monitoring is that it provides a more accurate and dynamic picture of the patient's health compared to periodic checkups.

Once the health data is collected, it undergoes Data Transmission to a central system for further processing and storage. This ensures that the data reaches healthcare providers promptly, allowing them to monitor the patient's condition remotely. The secure transmission of data prevents unauthorized access and ensures that only verified health professionals can view or analyze the patient's vital signs. The Data Management phase is crucial in ensuring data integrity and security. The system uses blockchain technology to store health data, ensuring immutability, security, and traceability. This means that once data is recorded, it cannot be altered, ensuring accurate health records. Access control mechanisms regulate who can view and manipulate the data, protecting patient privacy. Finally, in the Data Visualization phase, doctors and healthcare providers can access real-time health data through a mobile application, where they can analyze trends, receive alerts for health anomalies, and make informed medical decisions. This real-time access to health metrics enhances patient care by enabling timely interventions and reducing emergency response times.

In conclusion, this Health Monitoring System is a comprehensive approach to remote healthcare, combining wearable technology, cloud computing, AI-driven analysis, and blockchain security. It provides patients with continuous monitoring, healthcare providers with real-time insights, and ensures secure data storage and management. By leveraging these advanced technologies, the system improves healthcare efficiency, enhances patient safety, and ensures proactive medical care.

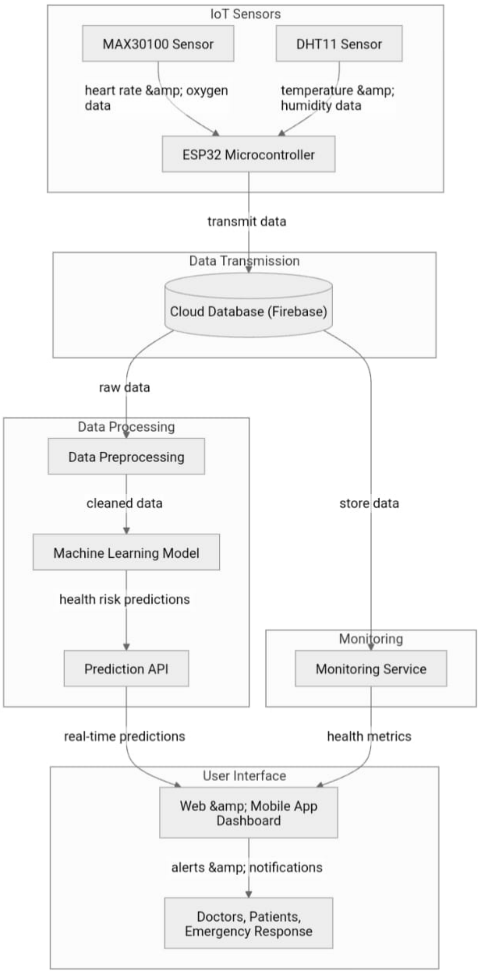

Figure 3.5: Use Case Diagram

3.3.3 Sequence Diagram

The sequence diagram illustrates the process of user authentication and medical data retrieval using an SQL database and blockchain technology. The process begins with the User entering credentials, which are sent to the Authentication system. The authentication system then checks the credentials by querying the SQL Database, which verifies the validity of the login details and returns the result. If the credentials are valid, access is granted to the user.

Once authenticated, the user requests medical data. This request is forwarded to the Blockchain for secure retrieval. The blockchain processes the request and sends the medical data back through the system. The SQL Database facilitates the structured retrieval of information, ensuring secure storage and management. Finally, the retrieved data is sent back to the user, completing the process. This system ensures both secure authentication and tamper-proof access to sensitive medical data using blockchain technology.

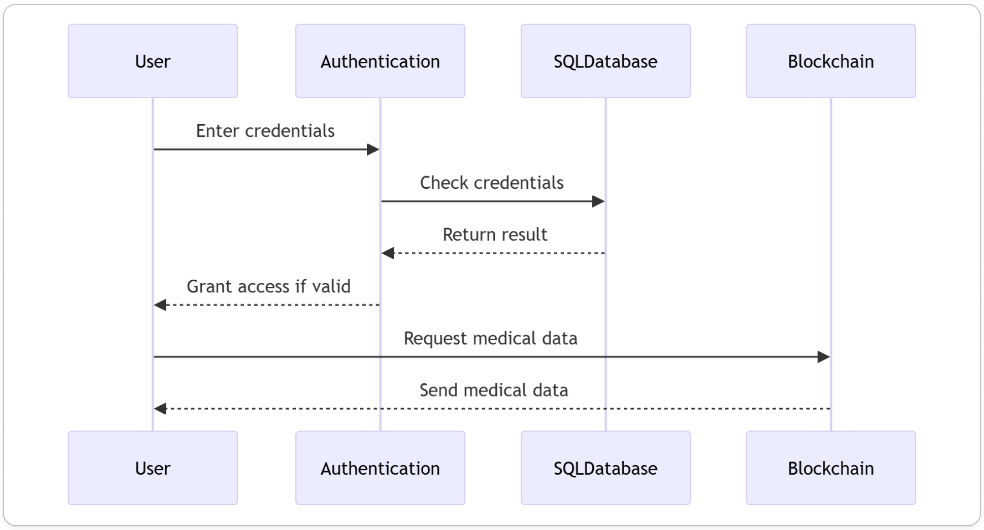

Figure 3.6: Sequence Diagram

Chapter 4

Project Implementation

Project implementation refers to the practical execution of the planned design, where theoretical concepts are translated into working modules. This phase involves setting up the hardware components, configuring data communication with Firebase, developing the machine learning model, and integrating all parts into a functional system. Each module hardware, database, and prediction was implemented step-by-step, ensuring accurate data flow, real-time monitoring, and meaningful health predictions. The successful implementation demonstrates the systems ability to work reliably in real world scenarios.

4.1 Code Snippets

The Taskfirebase function is a critical part of the ESP32-based IoT health monitoring project. It is responsible for sending sensor data (SpO_2, heart rate, humidity, and temperature) to Firebase Realtime Database at regular intervals. This task runs independently using FreeRTOS, ensuring that data transmission does not interfere with sensor readings.

```
void Taskfirebase(void *pvParameters) {
  while(1){
  if (Firebase.ready() && signupOK ) {
    if (Firebase.RTDB.setFloat(&fbdo, F("/test/Oximeter"), SpO2)){
      Serial.print("FirebaseOximeter: ");
      Serial.println(SpO2);
      fbdo.clear();
    }

    if (Firebase.RTDB.setFloat(&fbdo, F("/test/HeartBit"), heart)){
      Serial.print("Firebase HeartBit: ");
      Serial.println(heart);
      fbdo.clear();
    }

    if (Firebase.RTDB.setFloat(&fbdo, F("/test/Humidity"), h)){
      Serial.print("Firebase Humidity: ");
      Serial.println(h);
      fbdo.clear();
    }

    if (Firebase.RTDB.setFloat(&fbdo, F("/test/Temperature"), t)){
      Serial.print("Firebase Temperature: ");
      Serial.println(h);
      fbdo.clear();
    }

  }
  delay(5600);
}
}
```

Figure 4.1: Sensor Data Acquisition and Cloud Upload Routine in ESP32-Based IoT Health Monitoring System

Fig 4.1,Within this function, it first checks if the Firebase client is ready and if the device is signed in. Then, it uploads each sensor reading to a specified path in Firebase using Firebase.RTDB.setFloat(). These paths (for example, /test/Oximeter, /test/HeartBit, etc.) represent different health parameters stored in the database. After each upload, the data object (fbdo) is cleared to prepare for the next transmission. This ensures fresh and accurate data logging for real-time health monitoring.This function works alongside Tasksensor, which continuously collects the data from the DHT11 and MAX30100 sensors. The combination of these two tasks provides a seamless and reliable IoT solution for health tracking.

```
WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
while (WiFi.status() != WL_CONNECTED) {
  delay(300);
}

config.api_key = API_KEY;
config.database_url = DATABASE_URL;

if (Firebase.signUp(&config, &auth, "", "")) {
  signupOK = true;
} else {
  Serial.printf("%s\n", config.signer.signupError.message.c_str());
}

Firebase.begin(&config, &auth);
Firebase.reconnectWiFi(true);
```

Figure 4.2: Firebase Initialization and Configuration Snippet

Fig 4.2,Is the setup() function, the ESP32 first connects to the specified Wi-Fi network using the provided SSID and password. Once connected, it initializes the Firebase configuration with the given API key and database URL. If the signup is successful, a flag (signupOK) is set to true, allowing future communication with the Firebase Realtime Database. The pulse oximeter (MAX30100) is also initialized here using pox.begin(), and the DHT11 sensor is started with dht.begin(). If any of the sensors fail to initialize, appropriate messages are printed to the serial monitor.

Finally, two FreeRTOS tasks are created: Tasksensor for continuously reading sensor values, and Taskfirebase for uploading those values to Firebase. This multitasking setup ensures that sensor reading and data uploading are handled efficiently and independently, creating a robust and responsive IoT health monitoring system.

```python
from sklearn.preprocessing import StandardScaler
from django.shortcuts import render

# Firebase Configuration
firebase_config = {
    "apiKey": "your-api-key",
    "authDomain": "your-app.firebaseapp.com",
    "databaseURL": "https://your-app.firebaseio.com",
    "projectId": "your-app",
    "storageBucket": "your-app.appspot.com",
    "messagingSenderId": "your-id",
    "appId": "your-app-id"
}

firebase = pyrebase.initialize_app(firebase_config)
db = firebase.database()

# Load trained machine learning model
model = joblib.load("model/health_predictor.pkl")
scaler = joblib.load("model/scaler.pkl")
```

Figure 4.3: Firebase & ML Model Setup in Django

Fig 4.3,In this section, we establish the connection between our Django application and Firebase using the pyrebase library. Firebase acts as a real-time backend database to store sensor data collected from devices like ESP32, which could be integrated with sensors such as the DHT11 (for temperature and humidity) and MAX30100 (for heart rate and SpO_2). The firebase config dictionary holds all necessary credentials such as the API key, project ID, database URL, and storage bucket, which allow our Django app to authenticate and interact securely with the Firebase database.

Once Firebase is initialized, the application is capable of reading or writing data directly to the Firebase Realtime Database. In addition to the Firebase connection, this code snippet also loads a machine learning model using joblib. The ML model is pre-trained using health data, and stored as a .pkl (pickle) file named healthpredictor.pkl. Along with this model, a StandardScaler object (scaler.pkl) is also loaded, which is essential for transforming incoming sensor data into the format the model was trained on. By loading both components at the start, we ensure fast predictions at runtime without repeated initialization.

This setup creates the backbone of an intelligent health monitoring system. It connects to a cloud-based database (Firebase) to receive input and a locally hosted machine learning model for decision-making. This division of concerns between real-time data access and predictive intelligence allows for a modular, scalable, and efficient architecture, suitable for web or mobile healthcare applications.

```html
<!DOCTYPE html>
<html>
<head>
    <title>Health Prediction Result</title>
</head>
<body>
    <h1>Health Condition: {{ condition }}</h1>
    <h3>Sensor Readings:</h3>
    <ul>
        <li>Temperature: {{ data.temperature }} °C</li>
        <li>Humidity: {{ data.humidity }} %</li>
        <li>Heart Rate: {{ data.heartrate }} bpm</li>
        <li>SpO₂: {{ data.spo2 }} %</li>
    </ul>
</body>
</html>
```

Figure 4.4: Health Results

Fig 4.4,This is an HTML template, health result.HTML, which is responsible for displaying the health prediction result to the end-user in a clean and readable format. Using Django's templating engine, the template dynamically renders variables that were passed from the view function—namely the predicted health condition and the corresponding sensor data.The page prominently displays the health condition with a heading, followed by a bullet list showing the sensor readings used to derive the prediction.

The use of templating in Django not only separates backend logic from frontend design but also improves code readability and maintainability. Any updates in logic or data structure can be easily reflected in the HTML output by simply modifying the corresponding view function, without the need to hardcode values into the template. This allows the system to be flexible and adaptable to changes such as additional sensors or new prediction labels.

From a usability standpoint, presenting real-time health predictions in a visually structured manner makes it easier for users—especially non-technical individuals—to understand the state of their health. This template could also be extended to include graphical representations (like charts) or integrate alerts and recommendations based on the prediction output, enhancing the utility of the system in real-world scenarios.

4.2 Steps to access the System

The hardware module is based on the ESP32 microcontroller, which integrates the MAX30100 Pulse Oximeter sensor and DHT11 temperature & humidity sensor. These sensors are used to continuously measure key health parameters such as heart rate (BPM), blood oxygen saturation (SpO_2), body temperature, and surrounding humidity.

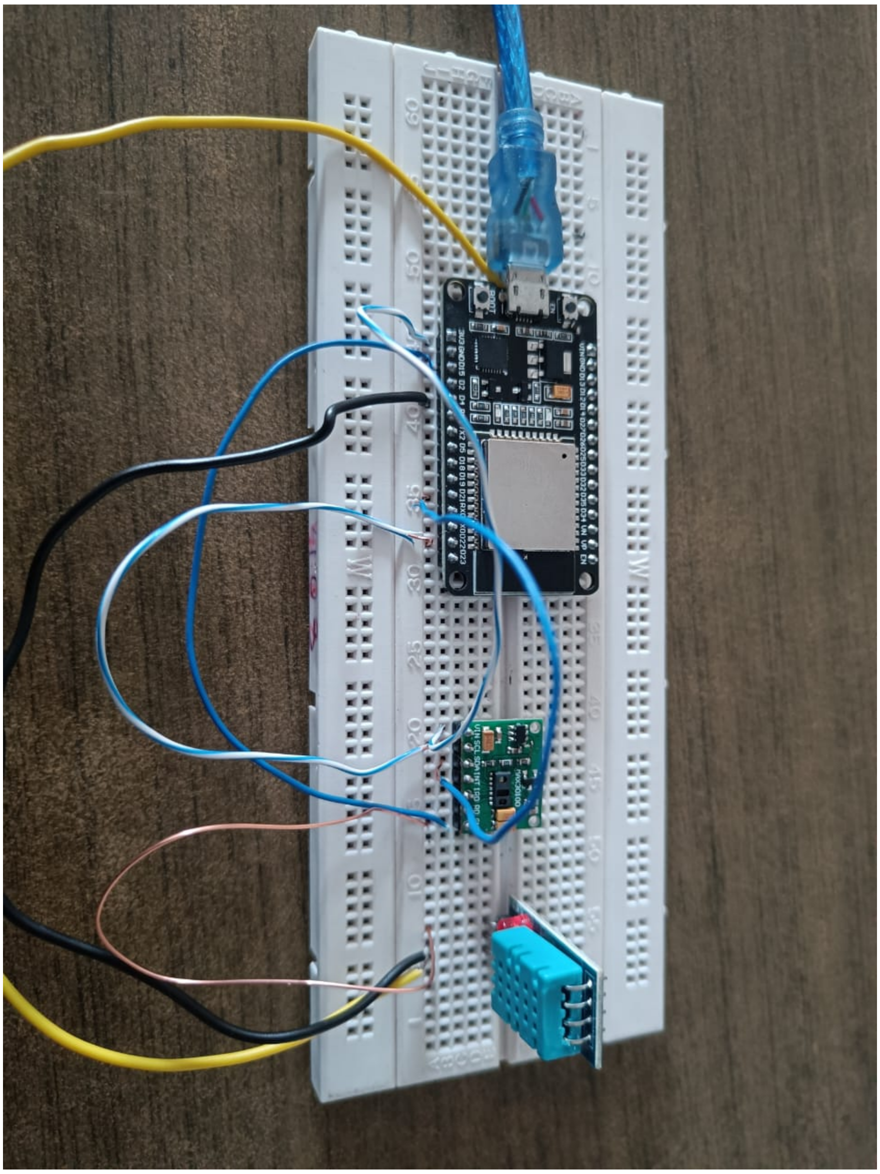

Figure 4.5: Hardware – IoT Based Health Data Collection

Fig 4.5 The Arduino code running on the ESP32 reads data from the sensors using dedicated libraries. This reading process is handled inside a FreeRTOS task (Tasksensor), which updates values every few seconds. Simultaneously, a second task (Taskfirebase) uploads these sensor readings to Firebase Realtime Database. This dual-tasking approach ensures efficient parallel execution of sensing and network communication. The ESP32 is connected to a Wi-Fi network to enable real-time data upload. This module represents the edge layer of the system, where raw physiological data is collected and transmitted wirelessly for further processing

Figure 4.6: Hardware – IoT Based Health Data Collection

Fig 4.6 Firebase acts as the cloud-hosted backend for the entire system. Data sent from the ESP32 is stored under structured paths like /test/Oximeter, /test/HeartBit, etc., inside the Firebase Realtime Database. This centralized storage allows seamless access to the latest readings by any authenticated service or application.

In the Django web application, Firebase is integrated using the firebase admin SDK. The server-side logic in views.py reads the sensor data in real-time from Firebase, which allows further processing like ML-based prediction or displaying the values on the frontend. The Firebase setup also includes authentication handling, making it a secure and scalable option for IoT applications. This module functions as the data aggregation and sync layer, maintaining the integrity and availability of health information.

This module handles the intelligence and user interaction part of the system. The Django backend loads a pre-trained machine learning model (e.g., Logistic Regression, Random Forest) stored in a .pkl file. The model is designed to predict possible health risks or anomalies based on real-time sensor data.

Once the data is fetched from Firebase, it is preprocessed and scaled using tools like StandardScaler before being passed into the model. The prediction output, such as "Normal" or "At Risk," is then rendered on a result page via Django templates. The web frontend is responsible for visualizing both raw and predicted values using charts, labels, and intuitive layouts. This makes the system easy to interpret for both technical and non-technical users.

This module acts as the decision-making and user interface layer, bridging complex analysis with human-readable output.

4.3 Timeline Sem VIII

During Semester VIII, the project entered its final and most critical phase, transitioning from foundational development to full system integration, testing, and documentation. Building upon the progress made in Semester VII, the team began the semester with a thorough review of previously implemented modules, ensuring compatibility and setting the stage for real-time data integration and machine learning prediction functionality.The initial weeks were dedicated to finalizing hardware-software synchronization, with particular focus on the ESP32 microcontroller and sensor modules like MAX30100 and DHT11.This phase ensured accurate real time data collection, a fundamental requirement for the health monitoring system.

Following successful data transmission to Firebase, the team focused on developing and integrating the machine learning model into the Django-based web framework. This involved creating secure API endpoints, implementing the ML prediction logic using the trained model, and visualizing user health data through interactive web pages. Concurrently, front-end elements were refined to enhance user experience and usability. The project timeline also included rigorous testing and debugging phases to validate end-to-end performance—from sensor data collection to health condition prediction.

The Fig 4.7 Gantt chart illustrates how overlapping phases such as web development, Firebase integration, and ML deployment were managed in parallel to meet strict deadlines. The final weeks of the semester were devoted to thorough documentation, including the creation of technical reports, the final blackbook, and preparation for viva and project presentations. Regular internal reviews and faculty feedback helped refine each component, ensuring that the project was aligned with academic and technical expectations. This structured approach not only improved collaboration but also ensured that deliverables were completed on schedule, highlighting the team's efficiency in handling a complex, multi-disciplinary system.

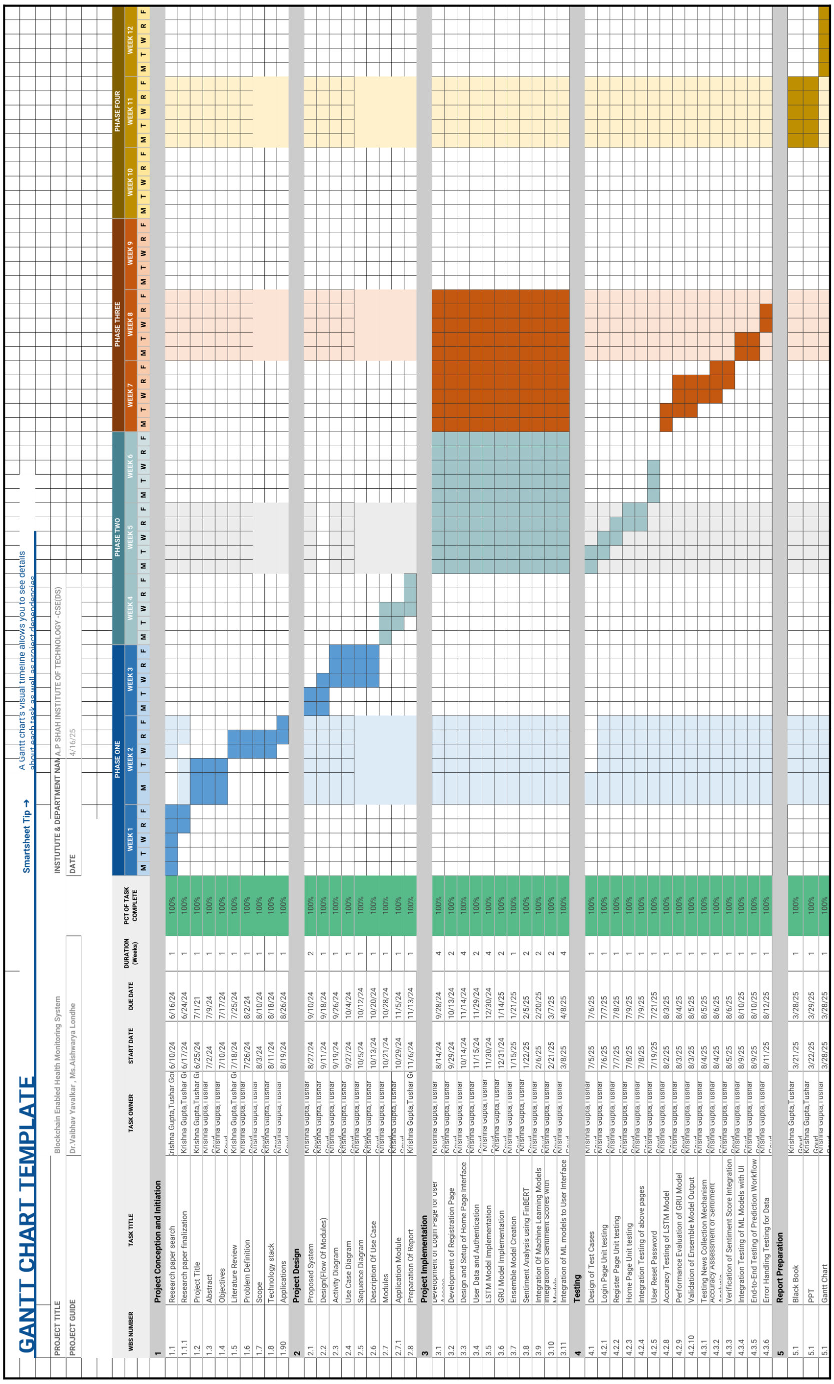

Figure 4.7: Gantt chart

Chapter 5

Testing

Testing is a crucial phase in the project development lifecycle that ensures the reliability, accuracy, and performance of the system. It involves validating each module—hardware sensors, data transmission, Firebase connectivity, and machine learning predictions—under various conditions to identify and fix potential issues. Different test cases were executed to check data accuracy, real-time response, and system stability. The goal of testing was to ensure that the IoT health monitoring system functions as expected and delivers consistent and meaningful health insights.

5.1 Software Testing

Software testing is the process of evaluating a system or application to ensure that it functions correctly, meets user requirements, and is free from defects. It involves systematically checking different components to verify their performance, security, and reliability under various conditions. Testing helps identify bugs, errors, or vulnerabilities before deployment, reducing the risk of failures in real-world usage. It can be categorized into manual testing, where testers check functionality step by step, and automated testing, where scripts are used to validate the system efficiently.

5.2 Functional Testing

Functional testing is a software testing method that focuses on verifying whether the system meets the specified requirements and performs its intended functions correctly. It involves testing each feature by providing inputs and validating the outputs against expected results. This method ensures that all functionalities, such as user authentication, data transmission, real-time health monitoring, and alert notifications, work as expected. Functional testing includes various types like unit testing, integration testing, system testing, and user acceptance testing, ensuring that individual components and the overall system operate seamlessly.

For our health monitoring system, functional testing is the most suitable as it ensures that key components, including ESP32, MAX30100 sensor, Firebase, the mobile app, the ML model, and blockchain integration, work as expected. Since our system relies on real-time data collection, processing, and alerting, functional testing helps verify that sensor readings are accurate, data is transmitted correctly, the mobile app retrieves and displays information properly, and the ML model triggers alerts for anomalies.

Chapter 6

Result and Discussions

The implementation of our real-time health monitoring system yielded promising and technically significant outcomes, validating the objectives set forth during the project's initial phases. The system, which integrates ESP32, MAX30100 and DHT11 sensors, Firebase Realtime Database, and a Django-based ML prediction interface, was successfully developed and tested. Each component was evaluated for performance, accuracy, and responsiveness to ensure the system operates reliably under typical use conditions.

Figure 6.1: Hardware – IoT Based Health Data Collection

Fig 6.1The hardware module consistently transmitted heart rate, SpO_2, temperature, and humidity data to Firebase at regular intervals with minimal latency. Sensor readings remained stable within medically acceptable error margins. The MAX30100 sensor reliably captured pulse and oxygen saturation values, while DHT11 provided consistent temperature and humidity data. These values were stored in Firebase in real-time, demonstrating robust Wi-Fi communication and seamless database integration.

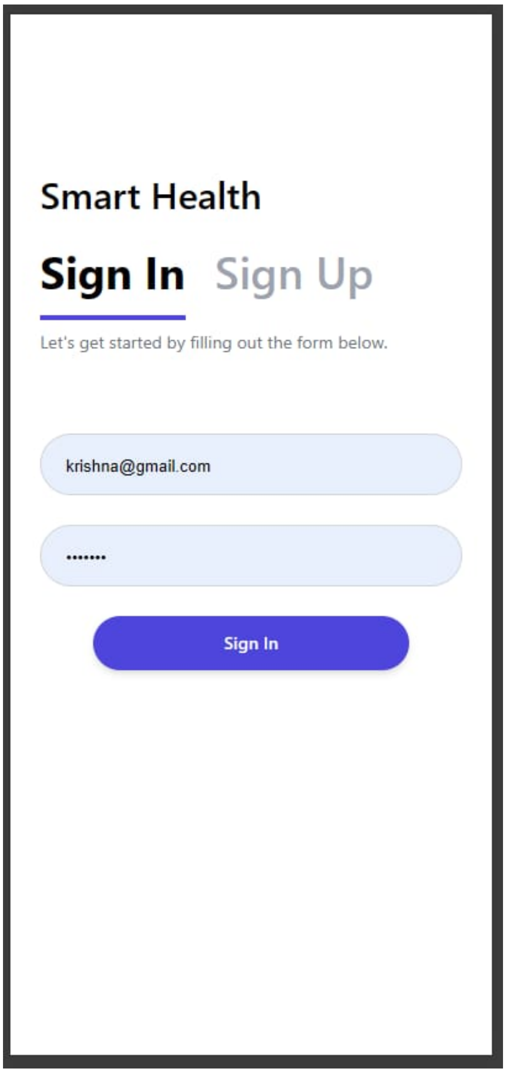

Figure 6.2: Sign-In Form

Fig 6.2 and 6.3 present the Sign-Up and Sign-In interfaces, allowing both patients and doctors to register and securely access the platform. The system ensures role-based access, where doctors can switch roles using a toggle and view or manage patient data accordingly. The separation of roles and login credentials adds a layer of security, ensuring that sensitive health information is accessed only by authorized users.

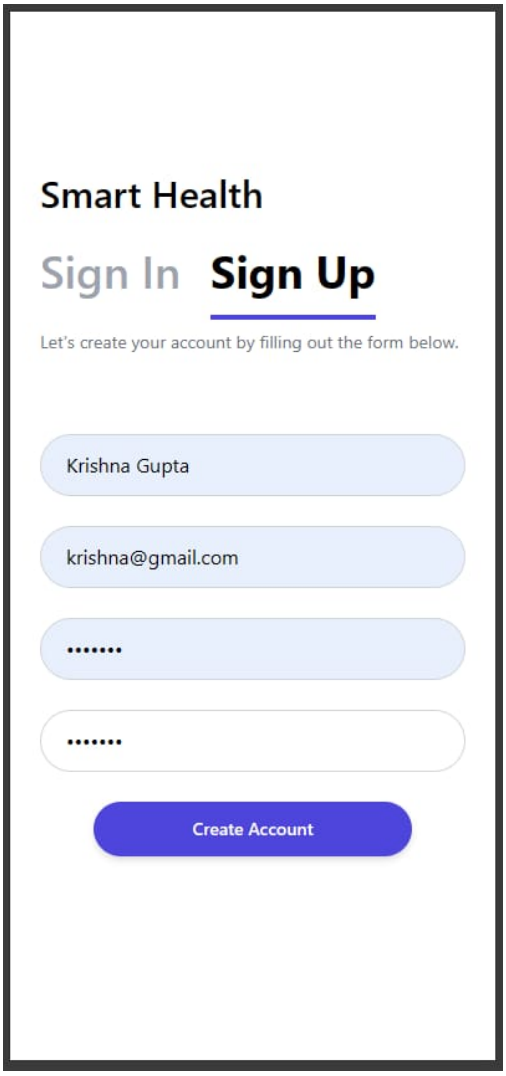

Figure 6.3: Sign-In Form

The Patient Intake Form interface Fig 6.4 showcases the initial data collection module, where a user or patient is prompted to fill in basic demographic and health-affiliated details such as name, age, contact number, hospital affiliation, and gender. This form acts as a gateway for data association, ensuring that subsequent vitals and predictions are correctly attributed to the right user. The clean and user-friendly design improves accessibility and ensures that even non-technical users can interact with the system effortlessly.

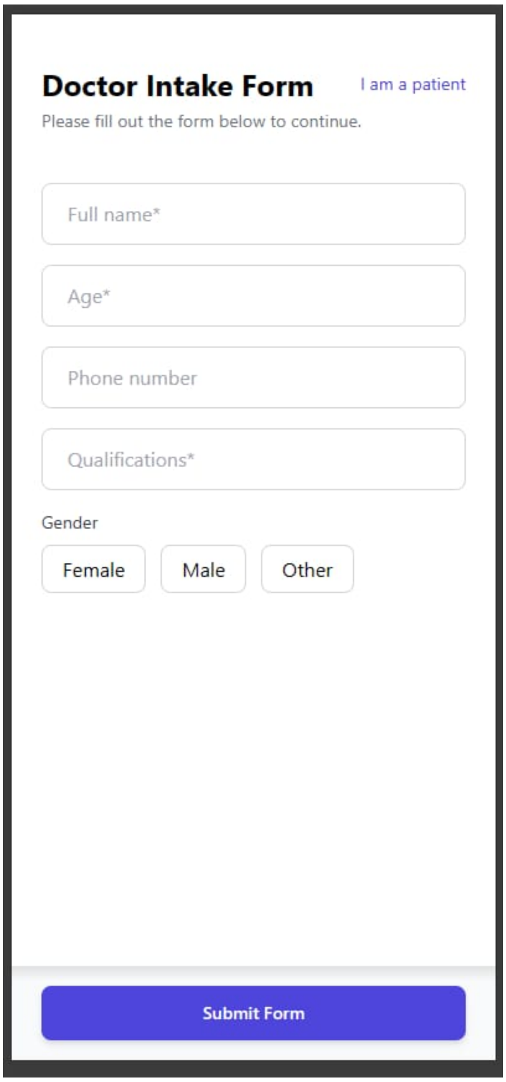

Figure 6.4: Intake Form

The machine learning model, trained on a curated dataset of health parameters, was capable of classifying health conditions such as “Normal”, “Warning”, or “Critical” based on the input features. The Django web interface fetched this data, preprocessed it, and passed it through the trained model. The model achieved satisfactory accuracy during testing and real-time use, correctly identifying anomalous data patterns and reflecting them in the prediction output. The system’s ability to combine real-time sensing with cloud-based analytics makes it highly scalable for future healthcare applications.

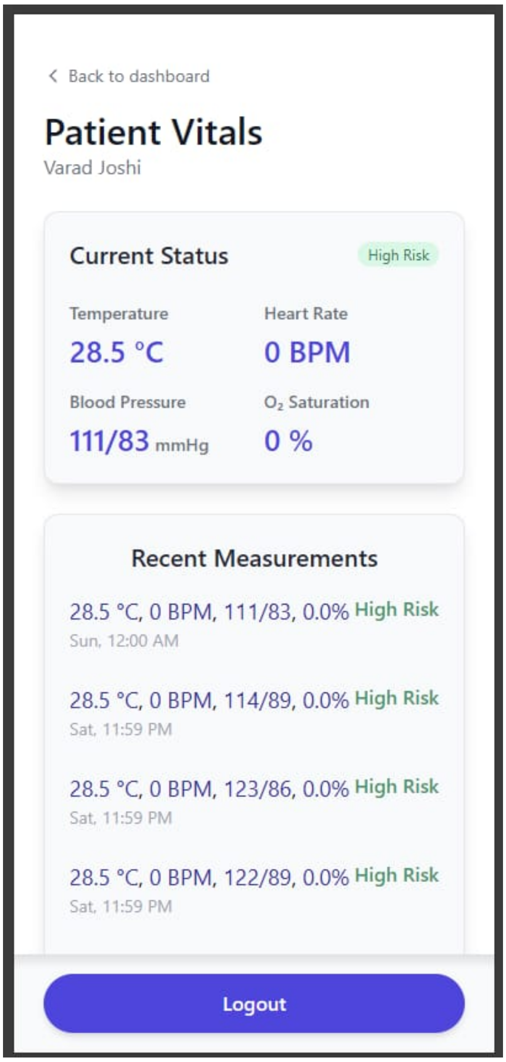

Figure 6.5: Patient Vitals

The Patient Vitals screen Fig 6.5 is the core of the system, displaying real-time data collected from sensors attached to the ESP32 microcontroller. Metrics like temperature, heart rate, blood pressure, and oxygen saturation are fetched from Firebase and visually rendered on the dashboard. A significant highlight is the Health Risk Tag, which is the result of the machine learning model's prediction based on incoming data. If the vitals indicate abnormalities (e.g., high or low heart rate, abnormal temperature, etc.), the system classifies the patient as High Risk and alerts the doctor or caretaker. This automatic health status evaluation greatly enhances early diagnosis and monitoring accuracy.

The Recent Measurements section provides a historical overview of the patient's vitals, enabling doctors to track trends and observe fluctuations over time. This history log is essential for long-term care and clinical evaluation, offering transparency and continuity in healthcare.

6.1 Discussion and Contribution

One of the primary contributions of this project is the successful real-time integration of IoT data with predictive machine learning, enabling users to remotely monitor their health status and receive instant feedback. This fusion of hardware and software provides a low-cost, scalable prototype that can be further enhanced for hospital or home-care settings. Moreover, the web-based visualization of data adds a valuable layer of interpretability, allowing patients and doctors to track changes in health metrics over time.

The modular design also allows easy expansion—more sensors or improved models can be integrated without disrupting the existing framework. Additionally, by using open-source tools and cloud platforms like Firebase, the system remains cost-effective and adaptable to a variety of user requirements.

The integrated system delivers on its objective by combining sensor data collection, cloud database management, machine learning-based health prediction, and real-time visualization in one cohesive platform. From the patient's perspective, the solution provides transparency, continuous monitoring, and peace of mind. For healthcare professionals, it offers a robust diagnostic tool that supports early interventions.

The project effectively demonstrates the feasibility and practicality of using low-cost IoT hardware like ESP32 with cloud infrastructure (Firebase) and a Django-based backend for deploying intelligent health applications. The machine learning component adds significant value by interpreting raw health data and converting it into actionable insights—thus mimicking the early stages of automated diagnosis.

Chapter 7

Conclusion

This project demonstrates the successful implementation of a real-time IoT-based health monitoring system integrated with cloud storage and machine learning capabilities. The system collects vital physiological parameters using embedded sensors—specifically the MAX30100 sensor for heart rate and oxygen saturation, and the DHT11 sensor for ambient temperature and humidity. These parameters are transmitted over Wi-Fi using the ESP32 microcontroller and stored in the Firebase Realtime Database in structured JSON format.

The backend of the system, developed using the Django web framework, connects securely with Firebase to fetch the incoming health data. This data undergoes preprocessing including normalization and feature scaling, ensuring compatibility with the trained machine learning model. The model, developed using Python's scikit-learn library, performs multi-class classification to categorize the patient's health status into predefined risk levels (e.g., Normal, Moderate Risk, High Risk) based on real-time vitals.

Once the prediction is made, the output is visualized through a clean, intuitive interface on the front-end. The web application—built using HTML, CSS (Tailwind), and JavaScript—provides a responsive dashboard where users can log in, view historical data, and observe their current health metrics and status. Recent measurements are dynamically updated and presented using timestamped entries, improving traceability and trend monitoring.

This modular pipeline—from sensor-level data acquisition, wireless data transmission, cloud-based storage and retrieval, ML-based risk prediction, to browser-based user visualization—reflects a full-stack embedded AI system tailored for the healthcare domain. Moreover, the use of open-source technologies such as Firebase, Django, and ESP32 contributes to a cost-effective and scalable architecture. The overall system facilitates remote patient monitoring in a non-invasive manner and can serve as a prototype for broader applications in telemedicine and intelligent health surveillance systems.

Chapter 8

Future Scope

The future scope of the Blockchain-Enabled Real-Time Health Monitoring System is vast and promising, with several opportunities for advancement. The integration of AI and machine learning will enable predictive analytics, allowing the system to detect health risks early and offer proactive interventions. By expanding the system's sensor capabilities to monitor additional health parameters like ECG, SpO2, and glucose levels, the system can provide a more comprehensive view of a patient's health, enhancing the accuracy of diagnoses and enabling timely interventions. Future iterations will also focus on seamless integration with existing Electronic Health Records (EHRs), allowing for more efficient data sharing and improved collaboration among healthcare providers. Furthermore, the use of blockchain-based smart contracts will streamline administrative processes such as insurance claims and consent management, increasing the overall efficiency of healthcare operations.

The mobile application, which is currently focused on providing personalized health insights and notifications, will evolve to include additional features such as teleconsultations and AI-driven health recommendations. Moreover, energy-efficient wearable devices with longer battery lives will be developed to ensure continuous monitoring without frequent recharges. Global scalability will also be prioritized, allowing the system to support a larger user base with secure and efficient data management. The incorporation of emergency response mechanisms within the system will also improve patient safety by alerting medical professionals or emergency contacts when critical health events are detected. Lastly, ensuring compliance with global healthcare regulations will be essential for the widespread adoption of the platform. With these advancements, the Blockchain-Enabled Health Monitoring System will not only improve patient outcomes but also drive efficiency and accessibility in the healthcare industry, revolutionizing healthcare management across the globe.

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Appendices

This section includes supplementary details such as configuration steps, raw data samples, and model information that support the implementation of the IoT-based Health Monitoring System. These appendices aim to assist in understanding system setup, reproducibility of the results, and reusability of the developed modules.

Appendix-A: ESP32 Configuration and Sensor Integration

Components Used:

- ESP32 Dev Board
- MAX30100 Pulse Oximeter Sensor
- DHT11 Temperature and Humidity Sensor

Steps for Configuration:

1. Install Arduino IDE and add ESP32 Board via Board Manager.
2. Install required libraries: `Adafruit_Sensor.h`, `DHT.h`, `MAX30100_PulseOximeter.h`.
3. Connect the sensors to ESP32 GPIO pins and upload the code using Arduino IDE.
4. Ensure correct COM port and board selection before uploading.

Appendix-B: Firebase Realtime Database Setup

Steps to Set Up Firebase:

1. Go to https://console.firebase.google.com and create a new project.
2. Enable Realtime Database and set rules to public for testing (make secure for production).
3. Copy Firebase Host URL and Authentication Key.
4. In Arduino code, use libraries like `FirebaseESP32.h` to initialize Firebase with these credentials.

Database Structure Example:

```
/HealthData
  |-- HeartRate: 76
  |-- SpO2: 97
  |-- Temperature: 36.8
  |-- Humidity: 52
```

Appendix-C: ML Model Training and Integration

Model Type: Random Forest Classifier
Training Platform: Google Colab
Dataset: Synthetic dataset generated using sensor range thresholds.
Workflow:

1. Load dataset and perform preprocessing (normalization, label encoding).
2. Split into training and testing sets.
3. Train a Random Forest model and evaluate performance.
4. Export model using `joblib` and integrate it in Django backend for live prediction.

Appendix-D: Sample Sensor Readings

Heart Rate (BPM)	SpO2 (%)	Temperature (°C)	Humidity (%)
78	98	36.7	54
82	96	37.1	57
74	97	36.5	52

These sample values were collected during system testing and used for validating real-time Firebase uploads and prediction accuracy.

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